

1-4-26. HW-4

Rout Delta

AME.

(191)

2) PDE : $\frac{\partial^2 T}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = 0$.

BC 1: $T(r, 0) = 0$

BC 2: $\frac{\partial T}{\partial z} \Big|_{z=\alpha} = 0 \iff \frac{\partial T}{\partial z} \Big|_{z=\alpha} = 0$.

BC 3: $\frac{\partial T}{\partial r} \Big|_{r=0} = 0$

BC 4: $\frac{\partial T}{\partial r} \Big|_{r=1} = f(z)$.

Also find $T(r, z)$ if $f(z) = 1$ (const function).

Solution

We solve this as an LSV problem.

$\therefore T(r, z) = R(r) Z(z)$

$\therefore \frac{\partial^2 (RZ)}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} (RZ) \right) = 0$.

$\frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} + \frac{1}{Rr} \frac{\partial}{\partial r} \left(r \frac{\partial R}{\partial r} \right) = 0$.

Let us use constant $-\lambda^2$

where $\boxed{\frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = -\lambda^2} \quad \& \quad \boxed{\frac{1}{R} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial R}{\partial r} \right) \right] = +\lambda^2}$
(i) (ii)

Equation (i) is easy.

$\frac{\partial^2 Z}{\partial z^2} + \lambda^2 Z = 0 \quad \therefore Z = A \cos(\lambda z) + B \sin(\lambda z)$

(*) If $\lambda = 0$, then $\frac{\partial^2 Z}{\partial z^2} = 0 \Rightarrow Z = Bz + A$.

BC 1: $T(r, 0) = Z(0) = 0 \Rightarrow \boxed{A = 0}$.

$$\psi = B \sin(\lambda z).$$

$$\text{BC 2: } \left. \frac{\partial \psi}{\partial z} \right|_{z=a} = 0.$$

$$\Rightarrow B \cos(\lambda a) = 0.$$

$$\therefore \lambda a = \left(n + \frac{1}{2}\right) \pi. \quad \text{("quantized" condition)}$$

$$\therefore \lambda_n = \left(n + \frac{1}{2}\right) \frac{\pi}{a}.$$

$$\therefore z_n \text{ (z eigenfunctions)} = B \sin\left(\left[n + \frac{1}{2}\right] \frac{\pi}{a} z\right).$$

Equation (ii)

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial R}{\partial r} \right) = \lambda_n^2 R.$$

$$\Rightarrow \frac{\partial R}{\partial r} + r \frac{\partial^2 R}{\partial r^2} = \lambda_n^2 r R.$$

$$\Rightarrow \boxed{r^2 \frac{\partial^2 R}{\partial r^2} + r \frac{\partial R}{\partial r} - \lambda_n^2 r^2 R = 0.}$$

$$\therefore R_n = A_n I_0(\lambda_n r) + B_n K_0(\lambda_n r)$$

$$\text{BC 3: } \left. \frac{\partial \psi}{\partial r} \right|_{r=0} = 0 \rightarrow R'_n(0) = 0.$$

$$\therefore R'_n = A_n I_1(\lambda_n r) (-\lambda_n) + (-\lambda_n) B_n K_1(\lambda_n r)$$

$$R'_n(0) = +\lambda_n A_n \underbrace{I_1(0)}_{\text{undefined}} - \lambda_n B_n \underbrace{K_1(0)}_{\text{undefined}}$$

$$\therefore B_n = 0.$$

$$\therefore R_n(0) = \boxed{A_n \lambda_n I_1(0) = 0} \rightarrow \text{satisfied}$$

$$\therefore R_n(r) = A_n I_0(\lambda_n r).$$

(Pg 3)

$$T(r, z) = R(r) \cdot Z(z).$$

$$= \sum_{n=0}^{\infty} A_n I_0(\lambda_n r) \cdot B \sin\left[\left(n+\frac{1}{2}\right)\frac{\pi}{a} z\right].$$

$$= \sum_{n=0}^{\infty} (A_n \cdot B) \cdot I_0(\lambda_n r) \cdot \sin\left[\left(n+\frac{1}{2}\right)\frac{\pi}{a} z\right].$$

$$T(r, z) = \sum_{n=0}^{\infty} C_n I_0(\lambda_n r) \cdot \sin\left[\left(n+\frac{1}{2}\right)\frac{\pi}{a} z\right].$$

Now we use BC 4.

$$BC4 \rightarrow \left. \frac{\partial T}{\partial z} \right|_{z=1} = f(z).$$

$$\Rightarrow \sum_{n=0}^{\infty} C_n(\lambda_n) I_1(\lambda_n) \sin\left[\left(n+\frac{1}{2}\right)\frac{\pi}{a} z\right] = f(z) \dots \text{(iii)}$$

We use the eigenvalue expansion on $f(z)$

$$\therefore f(z) = \sum_{n=0}^{\infty} K_n \cdot \sin\left[\left(n+\frac{1}{2}\right)\frac{\pi}{a} z\right] \dots \text{(iv)}$$

$\therefore$ For any arbitrary λ_k .

$$\rightarrow C_k \cdot \lambda_k \cdot I_1(\lambda_k) \sin(\lambda_k z) = K_k \sin(\lambda_k z) \left[\begin{array}{l} \text{from (iii)} \\ f(z) \text{ (iv)} \end{array} \right].$$

$$\therefore \boxed{K_k = C_k \cdot \lambda_k \cdot I_1(\lambda_k)} \dots \text{(v)}$$

Use (iv) further: —

$$f(z) = \sum_{n=0}^{\infty} K_n \underbrace{\left(\sin(\lambda_n z) \right)}_{\rightarrow \phi_n(z)} \quad \boxed{\text{from (v)}}$$

Now our selected eigenfunctions are orthogonal
i.e. $\boxed{\phi_n(z) \cdot \phi_m(z) = 0 \text{ if } (n \neq m)}$.

$\therefore$ multiplying $\phi_m(z)$ on both sides of (iv).

$$\therefore \phi_m(z) \cdot f(z) = \sum K_n \left[\underbrace{\phi_n(z)} \cdot \underbrace{\phi_m(z)} \right]$$

$\rightarrow$ Only nonzero terms that survive this due to orthogonality property is, when $\boxed{n=m}$.

$$\therefore \phi_m(z) \cdot f(z) = K_m \cdot \phi_m^2(z).$$

$$\therefore K_m = \frac{\int_0^{\alpha} f(z) \sin\left[\left(m+\frac{1}{2}\right)\frac{\pi}{\alpha} z\right] dz}{\int_0^{\alpha} \sin^2\left[\left(m+\frac{1}{2}\right)\frac{\pi}{\alpha} z\right] dz}.$$

Thus find the values of C_k in (v)

$$K_m = C_m \cdot \lambda_m I_1(\lambda_m). \quad \left\{ \begin{array}{l} k/m \text{ dummy} \\ \text{variable} \end{array} \right\}$$

$$\therefore C_m = \frac{1}{\lambda_m I_1(\lambda_m)} \frac{\int_0^{\alpha} f(z) \sin(\lambda_m z) dz}{\int_0^{\alpha} \sin^2(\lambda_m z) dz}.$$

with $\boxed{\lambda_m = \left(m+\frac{1}{2}\right)\frac{\pi}{\alpha}}$

we can in fact calculate the normalization constant. aka $\frac{1}{\int_0^{\alpha} \sin^2(\lambda_m z) dz}$.

$$\int_0^{\alpha} \sin^2(\lambda_m z) dz = \int_0^{\frac{\alpha}{2}} dz - \int_0^{\frac{\alpha}{2}} \frac{\cos(2\lambda_m z)}{2} dz$$

$$\int_0^a \sin^2(\lambda_m z) dz = \alpha/2 - \frac{1}{4\lambda_m} \left[\sin^2 \left(\frac{m\pi}{a} z + 2 \cdot \frac{1}{2} \frac{\pi}{a} z \right) \right]_0^a \quad \text{Fig. 5}$$

$$= \alpha/2 - \frac{1}{4\lambda_m} \left[\sin^2 \left(\frac{m\pi}{a} a + \frac{\pi a}{a} \right) - 0 \right]$$

$$= \alpha/2 - \frac{1}{4\lambda_m} \sin^2 \left[\frac{\pi a}{a} (2m+1) \right] = \alpha/2 - \frac{1}{4\lambda_m} \sin^2 \left[2\pi \left\{ \frac{\pi}{a} (m+\frac{1}{2}) \right\} \right]$$

$$\Rightarrow \alpha/2 - \frac{1}{4\lambda_m} \sin^2(2\pi m) = \alpha/2 - \frac{1}{4\lambda_m} \sin^2[2m\pi + \pi] = 0$$

$$\Rightarrow \text{Normalised factor: } \int_0^a \sin^2(\lambda_m z) dz = \alpha/2$$

$$\therefore c_m = \frac{1}{\lambda_m \cdot I_1(\lambda_m)} \left(\frac{2}{a} \right) \left[\int_0^a f(z) \sin(\lambda_m z) dz \right]$$

$$\text{AND } K_m = \frac{2}{a} \left\{ \int_0^a f(z) \cdot \sin(\lambda_m \cdot z) dz \right\}$$

We fix $f(z) = 1$.

$$\therefore K_m^* = \frac{2}{a} \left(\int_0^a \sin(\lambda_m z) dz \right)$$

$$= \frac{2}{a} \left[\frac{\cos(0) - \cos(\lambda_m a)}{\lambda_m} \right] = \frac{2}{a\lambda_m} [1 - \cos(\lambda_m a)]$$

$$\left[\frac{2}{a\lambda_m} \sin^2 \left(\frac{\lambda_m a}{2} \right) = \frac{2}{a\lambda_m} \sin^2 \left(\frac{(m+\frac{1}{2})\pi}{2} \cdot \frac{a}{a} \right) \right]$$

Check:

$$\cos(\lambda_m a) = \cos \left((m+\frac{1}{2}) \frac{\pi}{a} \cdot a \right) = \cos(m\pi + \pi/2) = -\sin(m\pi) = 0$$

$$\therefore k_m^* = \frac{2}{\alpha \lambda_m} = \frac{2}{\alpha \left[\left(m + \frac{1}{2} \right) \frac{\pi}{\alpha} \right]} = \frac{4}{(2m+1)\pi}$$

Fig. 8

$$\therefore C_m^* = \frac{1}{\lambda_m I_1(\lambda_m)} \left(\frac{2}{\alpha} \right) \left(\frac{1}{\lambda_m} \right) = \frac{2}{I_1(\lambda_m) \lambda_m^2 \alpha}$$

$$\therefore K_m^* = \frac{4}{(2m+1)\pi}$$

$$C_m^* = \frac{8\alpha}{I_1\left(\frac{(2m+1)\pi}{2\alpha}\right) (2m+1)^2 \pi^2}$$

$\therefore$ for $f(z) = 1$,

we have

$$T^*(r, z) = \sum_{m=0}^{\infty} \frac{8\alpha}{I_1\left[\frac{(2m+1)\pi}{2\alpha}\right] (2m+1)^2 \pi^2} \cdot I_0\left[\frac{(2m+1)\pi}{2\alpha} z\right] \sin\left[\frac{(2m+1)\pi}{2\alpha} r\right]$$